

INTRODUCTORY COURSE TO SEASONAL ADJUSTMENT



4 - Calendar effects correction

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Objectives of this segment

In this segment, we will see the different types of calendar effects and how to remove them from a series.

At the end, you will know how to :

- identify the different types of calendar effects
- explain the reasons why we remove them
- model a series correction
- read the available diagnostics in JDemetra+

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1. Why remove calendar effects ?

1.1 The calendar heterogeneity

1.2 The different calendar effects

1.3 Correction before comparison

2. How to remove the calendar effects ?

3. Available tools in JDemetra+

4. Conclusion

The calendar heterogeneity

The calendar is heterogeneous :

- trading days :
 - days usually worked, taking into account the company's sector. Most frequently, they are Mondays through Fridays when not bank holidays.
- week-ends
- bank holidays

For a given year as well as throughout the years, every month doesn't have the same number of days per day-type :

⇒ all months aren't "equal", even for a given month type

This causes **calendar effects**.

The month/quarter length effect

The *leap year* effect

Examples :

- the more trading days in a month, the higher the monthly production is expected to be
- number of trading days in France (non-bank holidays Mondays, Tuesdays, . . . , Fridays) per quarter :

Year	Q1	Q2	Q3	Q4
2010	63	62	65	64
2011	64	62	64	63
2012	65	60	64	64
2013	63	60	65	63
2014	63	60	64	64
2015	63	60	65	64
Average	63,50	60,67	64,50	63,67

The “type of day” effect

Examples :

- Sales in retrain trade are bigger on Saturdays than any other day
- average number of days per quarter in France, between 1940 and 2015

Trading day	Q1	Q2	Q3	Q4
Monday	12,57	11,02	12,86	12,72
Tuesday	12,74	12,80	12,88	12,71
Wednesday	12,74	12,82	12,86	12,74
Thursday	12,74	11,84	12,87	12,70
Friday	12,76	12,80	12,84	12,72
Saturday	12,76	12,80	12,84	12,70
Sunday	12,76	12,80	12,86	12,71

Bank holiday	Q1	Q2	Q3	Q4
Monday through Friday à vendredi	0,91	3,75	1,42	2,14
Saturday	0,14	0,20	0,29	0,43
Sunday	0,13	0,20	0,29	0,42

Bank holidays effects (french calendar)

Type of bank holiday		Examples	Effects
Moving holiday	Apart from Easter	– Pentecost Monday – Ascension Thursday – Easter Monday	Type of day
	Easter		Gradual effect
Set date : these bank holidays always occur on the same month but not necessarily on the same day		01/01, 01/05, 08/05, 14/07, 15/08, 01/11, 11/11, 25/12	Type of day

The gradual Easter effect

For some series, variations linked to Easter can be seen over a few days prior or following the d day : it's the *gradual Easter effect*.

Example : flower and chocolate sales rise significantly as Easter approaches

Correction before comparison, just like seasonal correction

Calendar effects must be removed in order to do :

- temporal comparisons
 - example : between months of different lengths
- comparisons between industries or fields
 - example : between trade and industry ("type of day" effect)
- geographical comparisons (example : between countries)
 - example : France and germany do not have the same number of trading days each month
 - different bank holidays
 - different main religions (orthodox vs christian calendar)

Sommaire

1. Why remove calendar effects ?

2. How to remove the calendar effects ?

2.1 Seasonality in the calendar

2.2 Econometrics to remove calendar effects

2.3 Different sets of trading day regressors

2.4 How to choose between all sets of regressors and to test for the choice ?

3. Available tools in JDemetra+

4. Conclusion

Calendar effects are partly seasonal

some of the calendar effects are seasonal :

- the number of trading days is nearly always smaller in February than in March
- some months have more bank holidays (thus fewer trading days) than others
 - example : May in France

⇒ some calendar effects are already taken into account during the seasonal adjustment

Two solutions to remove calendar effects

2 possible corrections :

- either simultaneously :
 - correcting for calendar effects
 - removing seasonal effects
 - or in two steps :
 - removing calendar effects during the pre-adjustment phase
 - then removing seasonal effects in the decomposition phase
- ⇒ the second solution is most often chosen in the mainstream algorithms

The econometric model

Hypothesis : the effect of a month's length or of a type of day for a given month (or quarter) is constant on the study period. We will use an econometric model to estimate an average effect via GLS.

A more simple method used before efficient econometric softwares is the application of a coefficient to each month (quarter) of the raw series :

$$\text{coefficient} = \frac{\text{average number of days for this type of month over a "long period"}}{\text{number of the month's trading days}}$$

Important : in the methods presented here, seasonality can evolve (via the use of moving averages) but calendar effects CAN'T. This might have consequences when the series is very long : calendar effects may be incorrectly estimated at the end of the series, which usually is the period of interest.

Hypotheses on the types of days

Construction of a set of “calendar effects” regressors, fitted to each series

To identify “significant” effects and to reduce the number of parameters to estimate with the regression, we must make hypotheses. That is, to establish similarities between certain types of days.

- Worked Mondays
- Worked Tuesdays
- ...
- Worked Sundays
- Bank holiday Mondays
- Bank holiday Tuesdays
- ...
- Bank holiday Sundays

Bank holiday days are usually treated as Sundays, and the other types of days are differentiated or considered similar, depending on the phenomenon measured by the series at hand.

Trading day regressors

Regressors characteristics :

- seasonally adjusted

in order not to weaken the seasonal signal before decomposing the series

- economically relevant : hypothesis are made on the “type of day” effects

This leads to several sets of regressors (explanatory variables). The user or the algorithm will select the one that best corrects each series.

The following slide shows all sets used at Insee. The variables are formulated in “contrasts” → explained later with the model.

Examples of regressors

Sets of regressors	Hypotheses	Reference (contrast)	Number of regressors
REG1	(Worked Monday = ... = Worked Friday) and (Saturday = Sunday = Bank holidays)	Saturday, Sunday and Bank holidays	1 + LPY
REG2	(Worked Monday = ... = Worked Friday), (Saturday) and (Sunday = Bank holidays)	Sunday and Bank holidays	2 + LPY
REG3	(Worked Monday), (Worked Tuesday = ... = Worked Friday) and (Saturday = Sunday = Bank holidays)	Saturday, dimanches et Bank holidays	3 + LPY
REG5	(Worked Monday), ..., (Worked Friday) and (Saturday = Sunday = Bank holidays)	Saturday, Sunday and Bank holidays	5 + LPY
REG6	(Worked Monday), ..., (Worked Friday), (Saturday) and (Sunday = Bank holidays)	Sunday and Bank holidays	6 + LPY
Leap Year (LPY)	Every day has the same effect	All days	1

The econometric model (1/4)

We'll use a linear multivariate model. The explanatory variables are the number of days of a given day type : that makes 7 variables.

$$X_t = \sum_{i=1}^7 \alpha_i N_{it} + \varepsilon_t$$

- N_{it} is the number of Mondays ($i = 1$), ..., Sundays ($i = 7$)
- Example : $N_{3,t=jan2007}$ is the number of Wednesdays in January 2007 (ie. 4 or 5)
- α_i is the effect of a type of day i (coefficient of the variable)

As $N_t = \sum_{i=1}^7 N_{it}$, the regressors are co-linear. That prevents the estimation of the coefficients α_i .

The econometric model (2/4)

Let's re-write the model, splitting in two α_i , the effect of a type of day i :

- the effect specific to a type of day i : β_i
- the average effect of a random day $\bar{\alpha}$, with $\bar{\alpha} = \frac{1}{7} \sum_{i=1}^7 \alpha_i$

We have : $\alpha_i = \beta_i + \bar{\alpha}$

When replacing α_i , the model becomes :

$$X_t = \sum_{i=1}^7 \beta_i N_{it} + \bar{\alpha} \sum_{i=1}^7 N_{it} + \varepsilon_t$$

And with $N_t = \sum_{i=1}^7 N_{it}$ the number of days of the month t :

$$X_t = \sum_{i=1}^7 \beta_i N_{it} + \bar{\alpha} N_t + \varepsilon_t$$

The econometric model (3/4)

By hypothesis, $\sum_{i=1}^7 \beta_i = 0$. Meaning, the day-specific effect is null over a week, just as seasonality is null over a year.

This allows to write β_i as a function of the other coefficients and to get rid of the co-linearity issue.

β_7 usually is the Sunday coefficient. We re-write it as :

$$\beta_7 = -\sum_{i=1}^6 \beta_i$$

After replacing the new expression into the equation, the model becomes :

$$X_t = \sum_{i=1}^6 \beta_i N_{it} - \sum_{i=1}^6 \beta_i N_{7t} + \bar{\alpha} N_t + \varepsilon_t$$

The econometric model (4/4)

The term $\bar{\alpha}N_t$ is purely seasonal except for February.

We will adjust it by subtracting its long term average :

$\bar{\alpha}N_t$ becomes $\bar{\alpha}(N_t - \bar{N}_t)$

For Februaries $\bar{N}_t = N_t$ with $\bar{\alpha}N_t = 28.25$

Hence, the final model is :

$$X_t = \sum_{i=1}^6 \beta_i (N_{it} - N_{7t}) + \bar{\alpha}(N_t - \bar{N}_t) + \varepsilon_t$$

Two terms appear :

- the specific effect of a type of day i as a contrast between the number of days i and the number of Sundays and bank holidays
- an effect of the month's length. Once seasonally adjusted, this term comes down to the leap year effect (frequently written LY in softwares)

A few questions to answer

Before choosing a set of regressors, one must answer the following questions :

- can the series present calendar effects (economic coherence) ?
- which days do we expect to have an impact on the series values ?
 - Trading days from Monday to Saturday ?
 - Trading days from Monday to Friday ?
 - Can all trading days have the same impact on the series values ?

For more detail on the construction of the regressors and the estimation of the models, see the related article in the Biblio repository.

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2. How to remove the calendar effects ?

3. Available tools in JDemetra+

3.1 JDemetra+ regressors

3.2 testing for the coefficients' significativity

3.3 Test of the gradual Easter effect

4. Conclusion

Available tools in JDemetra+

JDemetra+ offers two sets of built-in regressors :

- Trading Days (6 regressors)
 - All days from Monday to Saturday are considered different
- Working Days (1 regressor)
 - The difference is only made between week days (Monday = Tuesday = ... = Friday) and week-ends
- Leap year(1 regressor)

Beware : it is likely that all your national bank holidays will not be implemented by default in JDemetra+, but you can fix that by creating a new calendar corresponding to your country or importing your own regressors. That is what we will do during this course.

Two possible tests

In the diagnostics panel ("Pre-processing"), JD+ offers significance tests to evaluate the used set of regressors :

- a Fisher test of joint nullity of the coefficients (H_0 all coefficients are null and H_1 at least one of the coefficients isn't null)
 - can lead to removing the set of regressors from the Reg-ARIMA model
- for each regressor, a Student test of coefficient nullity (H_0 the coefficient is null and H_1 the coefficient isn't null)
 - can lead to change the set of regressors

Is there a gradual Easter effect ?

Does the series present a gradual Easter effect ?

- If so, an extra regressor is added to remove the gradual effect
 - Variable “Easter” in JDemetra+
- If not, no variable is added
- JDemetra+ can “pre-test” the presence of a gradual Easter effect. If the test is positive, JD+ adds the regressor “Easter [n]” ($n=1, \dots, 20$) to the model.

In the diagnostics panel (“Pre-processing”), JDemetra+ offers a significance test of the coefficient régresseur “Easter [n]” (test de Student : H_0 $\text{coeff}(\text{Easter [n]}) = 0$ et H_1 $\text{coeff}(\text{Easter [n]}) \neq 0$)

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The essentials

- It is paramount to remove calendar effects before making temporal and spatial comparisons (mécanisme identique to seasonal adjustment, with the estimated effects being fixed for the whole study period)
- There are three types of calendar effects : the effect of month length, type of day and movable feast (Easter)
- We only remove non-seasonal calendar effects in order to preserve the raw signal seasonality for the decomposition step (x11).
- Custom sets of regressors can be calculated after making hypotheses on the likeness of day types
- However, creating regressors ex-nihilo is a tedious task (contrasts calculus and seasonal adjusting over a long period) : Insee supplies regressors for French series
- JDemetra+ tests the presence of residual calendar effects (called residual trading days effects in the software)